TurboRUL

AI predictive maintenance for aircraft engines — progress report
Phases 1–5 of 6 complete · 12 October 2026

1. Summary

TurboRUL predicts the **Remaining Useful Life (RUL)** of turbofan engines, in flight cycles, from sensor histories; attaches a calibrated 90% range, a risk tier and the sensors that drove each prediction; and serves this through a REST API and a three-screen planning dashboard. It uses NASA's public C-MAPSS dataset (simulated). Since the Phase 3 report, tuning moved to engine-level 5-fold cross-validation, all four subsets were trained and scored, and the uncertainty, explanation, API and dashboard layers were built.

- **All four subsets are covered.** Deployed model (LSTM, chosen on validation) test RMSE: FD001 15.2, FD002 24.5, FD003 14.9, FD004 24.6 cycles (five-seed means). On FD001 the best model is the 1D-CNN at 14.63; FD002/FD004 errors are dominated by engines whose true RUL exceeds the cap (Section 4).
- **Critical engines are caught.** Across all 707 test engines, the deployed models flag 92% of truly critical engines as Critical, and 100% when flagging on the lower end of the 90% range. No truly critical engines are labelled Healthy.
- **Ranges are close to, but below, their 90% target.** The deployed models' calibrated ranges cover FD001 82%, FD002 92%, FD003 89%, FD004 92% of test engines against capped truth, and less against uncapped truth; calibration rests on only ~20 validation engines per subset, so this is the main weakness to fix (Section 5).
- **Every prediction is explained** in cycles per sensor (TreeSHAP or integrated gradients). Fleet-level drivers are the temperatures, pressures, bypass ratio and shaft speeds that track compressor degradation (Section 6).
- **The product works end to end.** FastAPI service and Streamlit dashboard (fleet, engine detail, model comparison) run locally; a prediction with explanation takes 56 ms (target < 2 s) and a visitor's dashboard loads in 0.6 s (target < 3 s). 64 automated tests pass.

2. Objectives and status

Requirement	Status after Phase 5
FR-1 – FR-6 · data, preprocessing, split, baselines, deep models, evaluation	**Done** (Phases 1–3); now on all four subsets.
FR-7 · predict any engine via a function and REST endpoint	**Done.** *predict.py*; *POST /predict*, *GET /fleet/{subset}* and more.
FR-8 · uncertainty range per prediction	**Done.** MC dropout / quantile XGBoost, conformally calibrated to 90%.
FR-9 · risk tier from configurable thresholds	**Done.** Critical ≤ 30, Watch ≤ 60, Healthy otherwise; plain-language action.
FR-10 · top contributing sensors per prediction	**Done.** TreeSHAP (XGBoost), integrated gradients (LSTM/CNN), exact (linear).
FR-11 · dashboard: fleet list, engine detail, model comparison	**Done**, plus an About panel stating the data is simulated.
FR-12 / FR-13 · replay, fault-mode classification (P2)	Not started (optional, Phase 6).
Metric · FD001 test RMSE ≤ 15 (stretch ≤ 13)	**Partly.** Deployed LSTM 15.20; 1D-CNN 14.63 and XGBoost 15.03. Stretch not met.
Metric · ≥ 25% lower RMSE than linear	**Not met** on FD001 (8% for the deployed model); met on FD002 (16%) and FD004 (18%).
Metric · NASA score reported, beats baselines	**Mostly.** Every model beats linear on FD002/FD004 by a wide margin; on FD001 XGBoost and on FD003 the LSTM score slightly worse than linear (Table 2).
Metric · critical-engine recall ≥ 90%	**Met with the lower bound** (100% all subsets); 92% on the point prediction.
Metric · RMSE on all four subsets	**Met** (Table 2).
Metric · prediction < 2 s; dashboard load < 3 s	**Met.** Prediction 64 ms worst case (warm). Dashboard 0.6 s per visitor; 7.7 s once after a server start (Python imports).

3. What changed in Phases 4–5

- **Cross-validated tuning.** As planned in the Phase 3 report, Optuna now scores each trial by the mean RMSE over five engine-level folds (test-like truncated validation engines in each), pruning weak trials after any fold. This was fixed before any FD002–FD004 test score existed. The test set is still never loaded during tuning.
- **Compute compromise.** The LSTM is the slowest model to tune (about four CPU-hours on FD004), so it was tuned on FD001 only and its configuration reused on FD002–FD004 (marked *transfer*). XGBoost and the 1D-CNN were tuned on every subset. Deep trials trained on every second window (overlapping windows are near-duplicates).
- **Selection rule.** The final configuration of each family is the cross-validated (or transferred) one. The *deployed* family per subset is the one with the lowest mean validation RMSE over five seeds; test scores play no part in either choice.
- **Uncertainty.** Each model's own spread — MC dropout (50 passes) for LSTM/CNN, 5th/95th-percentile quantile XGBoost, a constant width for linear — is scaled by a split-conformal factor fitted on validation engines so that 90% of them fall inside. Validation truth is capped at the model's RUL cap for calibration (Section 5).

Subset	Model	Window / cap	CV RMSE (5 folds)	Trials (pruned)	Tuning time
FD001	XGBoost	50 / 140	14.31	30 (7)	32 min
FD001	1D-CNN	50 / 130	15.41	20 (5)	25 min
FD001	LSTM	50 / 130	13.87	20 (6)	62 min
FD002	XGBoost	50 / 140	17.11	30 (12)	33 min
FD002	1D-CNN	50 / 125	16.98	20 (5)	49 min
FD002	LSTM	50 / 130	13.87 (on FD001)	transferred	—
FD003	XGBoost	50 / 135	15.35	30 (11)	14 min
FD003	1D-CNN	50 / 125	15.99	20 (4)	44 min
FD003	LSTM	50 / 130	13.87 (on FD001)	transferred	—
FD004	XGBoost	50 / 135	17.33	30 (8)	36 min
FD004	1D-CNN	50 / 130	17.33	20 (4)	61 min
FD004	LSTM	50 / 130	13.87 (on FD001)	transferred	—

4. Results on all four subsets

Final configurations scored on each official test set; mean ± std over five seeds. ◆ marks the deployed model (lowest validation RMSE). Coverage and width describe the calibrated 90% ranges (uncapped truth).

Subset	Model	Params	W / cap	Test RMSE	RMSE vs capped truth	NASA score	Critical recall	Coverage	Mean width
FD001	Linear regression	default	30 / 125	16.47	15.10	413	60%	91%	59
FD001	XGBoost	tuned-cv	50 / 140	15.03 ± 0.09	15.00	441 ± 18	85%	93%	60
FD001	**LSTM ◆**	tuned-cv	50 / 130	**15.20 ± 0.59**	14.67	393 ± 41	85%	79%	38
FD001	1D-CNN	tuned-cv	50 / 130	14.63 ± 0.25	13.91	324 ± 25	94%	87%	42
FD002	Linear regression	default	30 / 125	29.14	16.42	12600	82%	76%	52
FD002	XGBoost	tuned-cv	50 / 140	24.09 ± 0.05	16.42	3920 ± 34	95%	78%	50
FD002	**LSTM ◆**	transfer	50 / 130	**24.47 ± 0.57**	14.15	4890 ± 573	98%	78%	55
FD002	1D-CNN	tuned-cv	50 / 125	25.76 ± 0.36	13.41	5955 ± 487	98%	74%	45
FD003	Linear regression	default	30 / 125	17.51	15.69	492	80%	85%	53

Subset	Model	Params	W / cap	Test RMSE	RMSE vs capped truth	NASA score	Critical recall	Coverage	Mean width
FD003	XGBoost	tuned-cv	50 / 135	15.00 ± 0.19	14.50	451 ± 16	89%	90%	43
FD003	**LSTM ◆**	transfer	50 / 130	**14.88 ± 0.77**	14.10	511 ± 262	96%	86%	44
FD003	1D-CNN	tuned-cv	50 / 125	14.96 ± 0.72	13.05	443 ± 94	96%	80%	39
FD004	Linear regression	default	30 / 125	30.21	18.40	7394	68%	71%	56
FD004	XGBoost	tuned-cv	50 / 135	25.80 ± 0.16	16.66	4583 ± 462	94%	67%	43
FD004	**LSTM ◆**	transfer	50 / 130	**24.64 ± 0.52**	14.17	3868 ± 367	83%	75%	56
FD004	1D-CNN	tuned-cv	50 / 130	27.44 ± 1.84	17.33	6389 ± 2175	88%	72%	73

Table 2. Final configurations per subset. W / cap: window length and RUL cap (cycles).

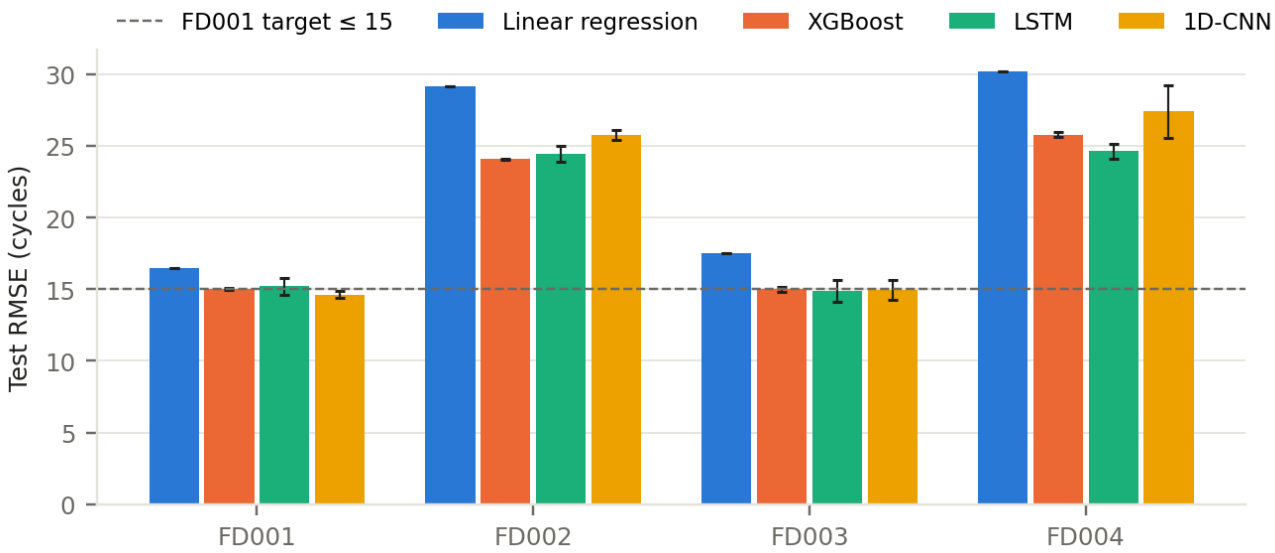


Figure 1. Test RMSE by subset and model (mean ± std over five seeds).

4.1 FD001: did cross-validated tuning help?

Model	Phase 3: single-split tuning	Phase 4: 5-fold CV tuning	Change
XGBoost	14.19 ± 0.23 (val 15.0)	15.03 ± 0.09 (val 14.2)	+0.84
LSTM	14.99 ± 0.67 (val 15.3)	15.20 ± 0.59 (val 13.7)	+0.21
1D-CNN	14.90 ± 0.55 (val 16.4)	14.63 ± 0.25 (val 15.4)	-0.27

Cross-validation did not improve FD001 test scores: XGBoost and the LSTM got slightly worse and the CNN slightly better. Its value is a more trustworthy validation estimate (five folds instead of 20 engines), not a better FD001 number. FD001 is no longer a clean hold-out for this comparison, so it is descriptive only; the protocol was fixed in advance and applied unchanged to FD002–FD004.

Why FD002 and FD004 look so much worse. Their test sets contain many engines with true RUL far above the cap (up to 194 cycles). A capped model predicts at most the cap for these, so their errors of 40–60 cycles dominate RMSE. Against truth capped at the model's cap (column *RMSE vs capped truth*), FD002 and FD004 are in line with validation. Per-condition normalisation works as intended: the operating-condition clusters are as tight on test engines as on training engines.

4.2 Where the errors are

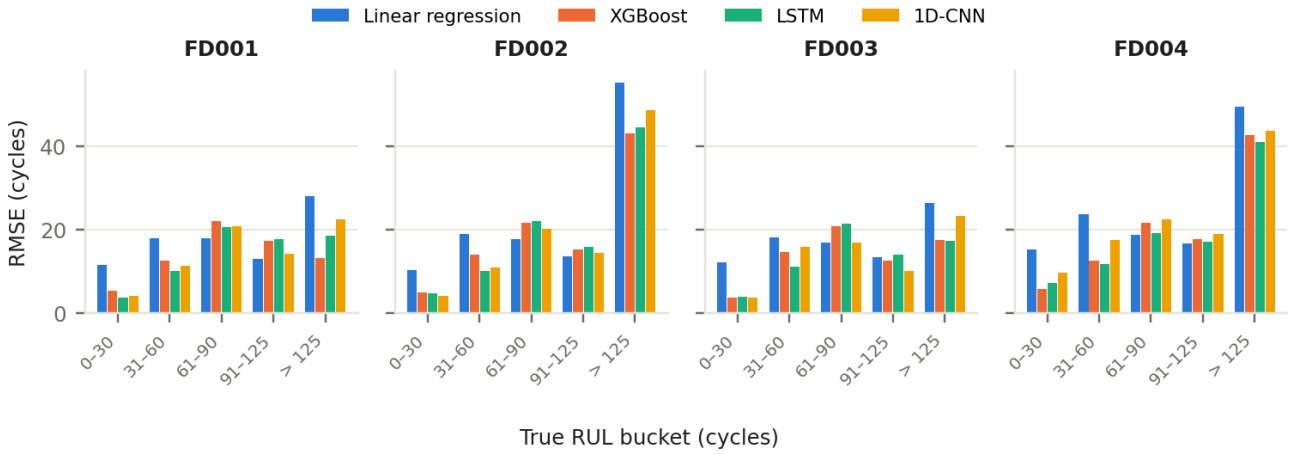


Figure 2. Test RMSE by true-RUL bucket, final configurations, mean over five seeds.

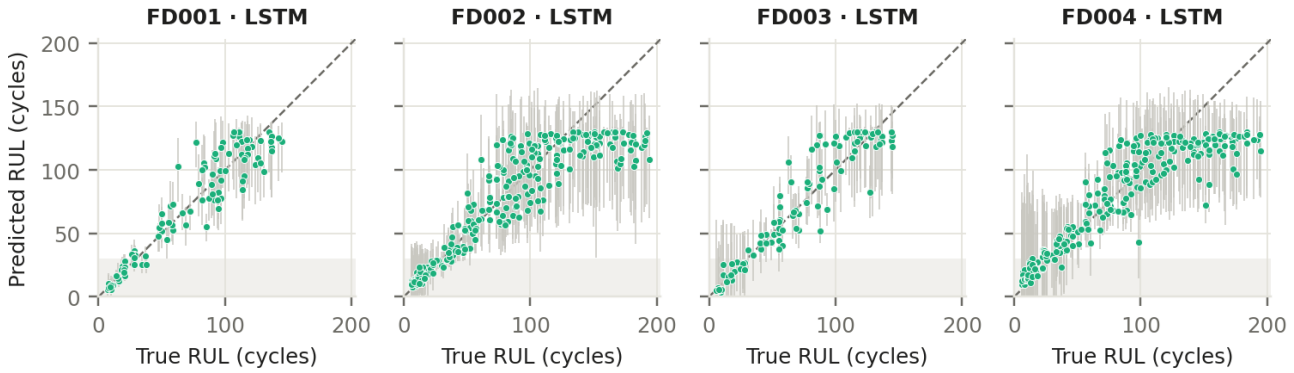


Figure 3. Predicted vs true RUL with calibrated 90% ranges, deployed model per subset (seed 0). Shaded: Critical band; dashed: perfect prediction.

5. Uncertainty and risk tiers

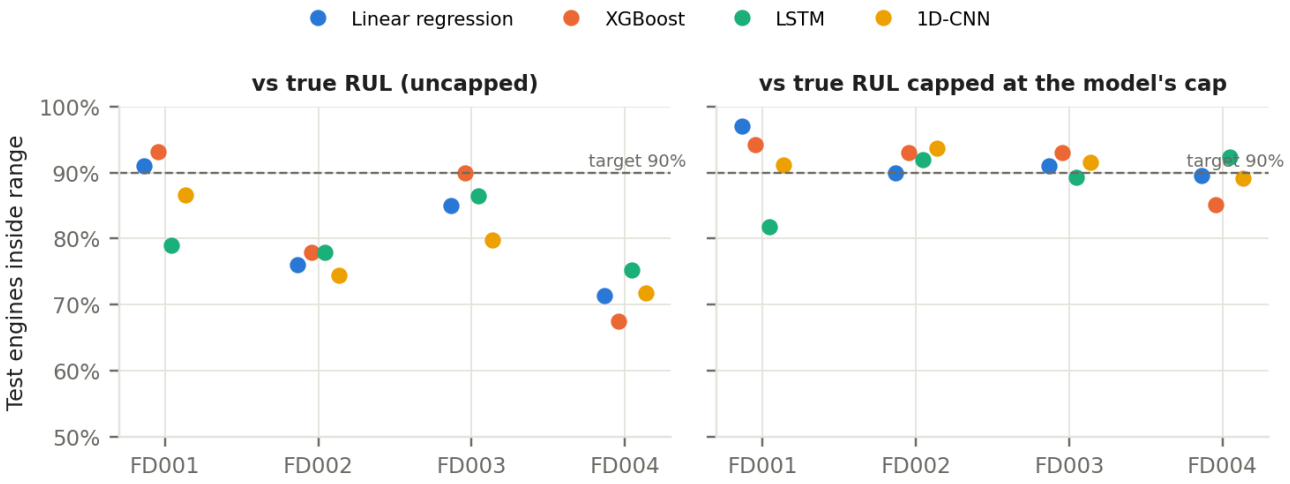


Figure 4. Share of test engines whose true RUL lies inside the calibrated 90% range. Left: uncapped truth. Right: truth capped at the model's RUL cap.

- **Coverage falls short of 90%, mildly.** Against capped truth it is FD001 82%, FD002 92%, FD003 89%, FD004 92% for the deployed models. The conformal factor is fitted on 10 cuts of each of ~20 validation engines, so it rests on few independent engines and is noisy. Calibrating on out-of-fold predictions for every training engine is the planned fix; it was not applied here, to avoid changing the method after seeing test coverage.
- **Uncapped coverage is lower on FD002 and FD004** because about a quarter of their test engines have true RUL above 125 cycles, which a capped model reports as "at least the cap". For planning this is harmless: those engines are Healthy under either reading.
- **Widths vary by engine.** On FD001 the deployed model's ranges are narrow near failure (median width 15 cycles for Critical engines) and wide for healthy ones, which is what a planner needs. On FD002–FD004 they stay wide even near failure, reflecting the noisier multi-condition data and the transferred LSTM configuration.

Risk tiers on the test fleets (deployed models, all four subsets):

True tier ↓ / predicted →	Critical	Watch	Healthy	Engines
Critical	146	13	0	159
Watch	10	81	23	114
Healthy	0	10	424	434

Critical precision is 94% and recall 92% on the point prediction. Flagging on the lower bound of the range raises recall to 100% at the cost of more Watch engines being raised early — the conservative choice for safety. The dashboard's recommended action already quotes the lower bound.

6. Explainability

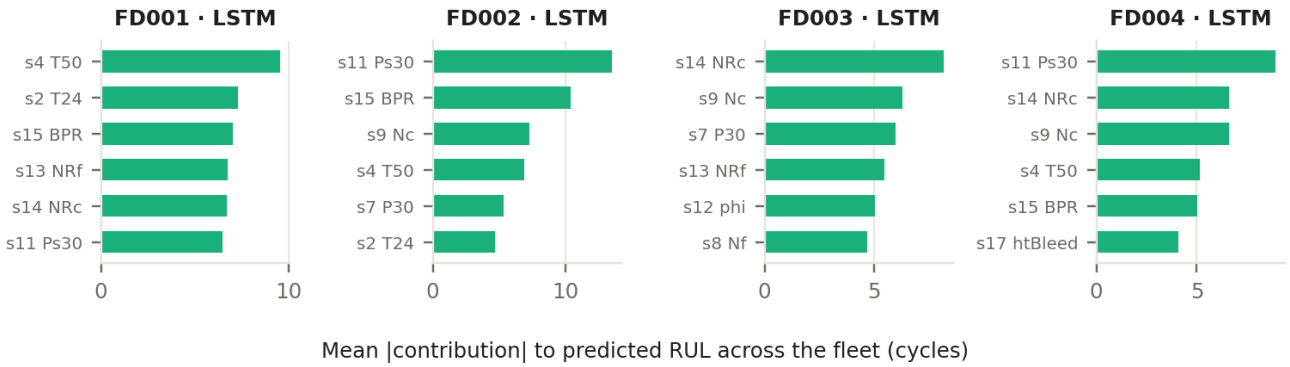


Figure 5. Fleet-level drivers: mean absolute contribution of each sensor to predicted RUL, deployed model per subset.

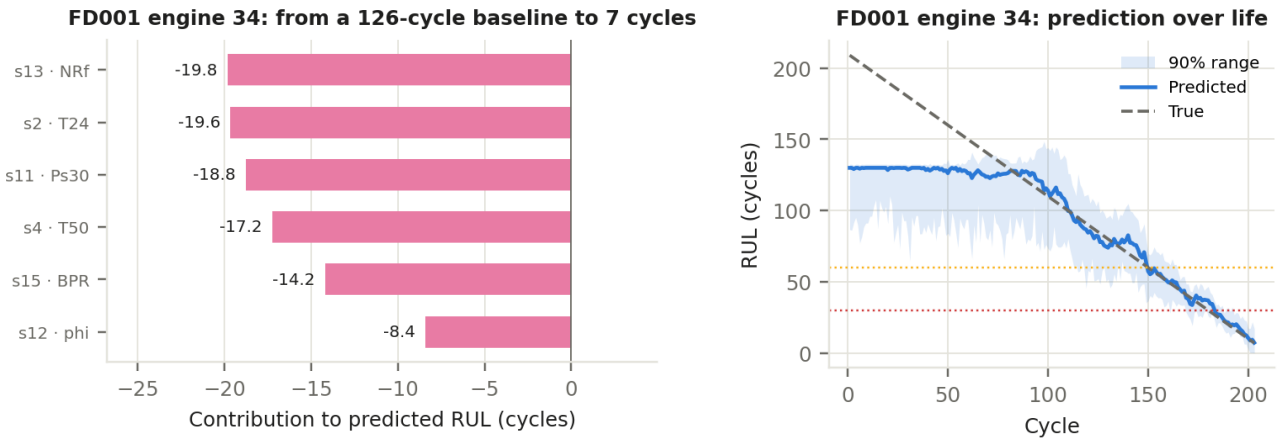


Figure 6. One Critical engine (true RUL 7). Left: per-sensor contributions from the average-engine baseline (magenta = towards failure). Right: the prediction as it would have evolved live, with its 90% range and the true RUL.

- The fleet-level drivers are the gas-path temperatures (s4 T50, s2 T24), HPC outlet static pressure (s11 Ps30), bypass ratio (s15) and shaft speeds (s9/s14 core, s13 fan): the same sensors the Phase 1 analysis found trending towards failure. On FD003, which adds a fan fault, shaft speeds lead.
- Contributions are additive in cycles (base value + contributions = raw prediction; exact for linear and XGBoost, within about 1–3 cycles for the integrated-gradients approximation), so a planner can read them directly. Correlated sensors share credit, so individual bars should not be over-interpreted.

7. Product: API and dashboard

Component	What it does
models/<subset>/	Exported bundle per subset: model, fitted preprocessor, metadata (params, calibration, test metrics). FD001: LSTM, FD002: LSTM, FD003: LSTM, FD004: LSTM.
predict.py	*predict(history)* → RUL, 90% range, tier, action, top sensors; *predict_trajectory* for the prediction at every past cycle.
api/main.py (FastAPI)	*POST /predict* (raw history in, prediction out, validated), *GET /fleet/{subset}* (filter by tier), */engines/{subset}/{unit}*, */fleet/{subset}/importance*, */results*, */subsets*, */health*; OpenAPI docs at */docs*.
app/dashboard.py (Streamlit)	Fleet overview, engine detail, model comparison and About. Talks to the API when *TURBORUL_API_URL* is set, otherwise calls the same service in-process (for single-process free hosting in Phase 6).

Latency (laptop CPU): a single prediction with explanation takes 60 ms on FD001 and 56 ms on FD004 (median; worst 79 ms). Scoring and explaining a whole fleet takes 0.5 s (FD001, 100 engines) and 1.1 s (FD004, 248 engines); the export step stores these results with each bundle, so the dashboard does not recompute them. A visitor's dashboard loads in 0.6 s; the first load after a server start takes 7.7 s, mostly importing PyTorch and scikit-learn.

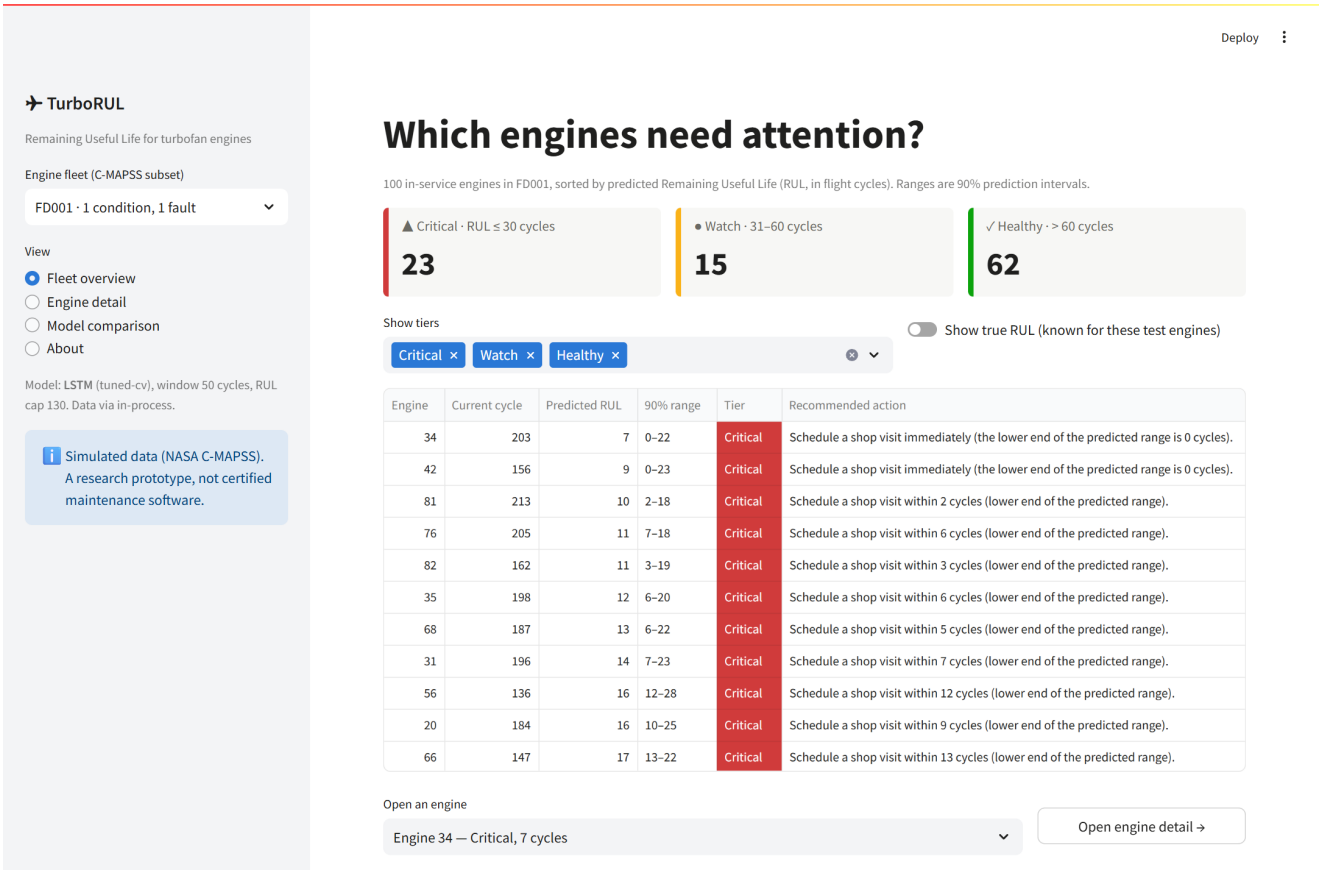


Figure 7. Fleet overview: tier counts, engines sorted by predicted RUL with 90% ranges and actions.

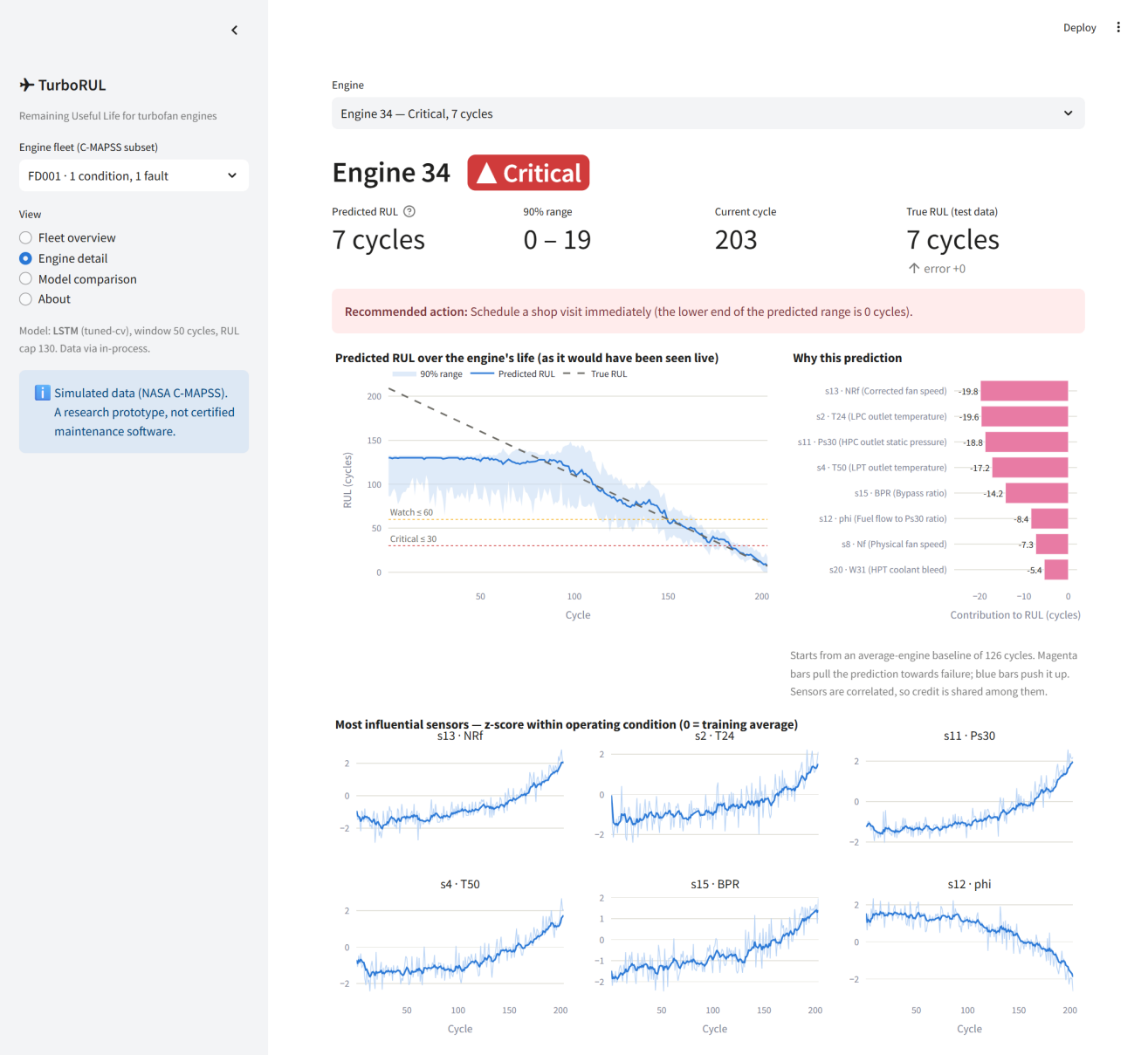


Figure 8. Engine detail: prediction, range, tier and action; RUL over life vs truth; why-this-prediction; sensor trends.

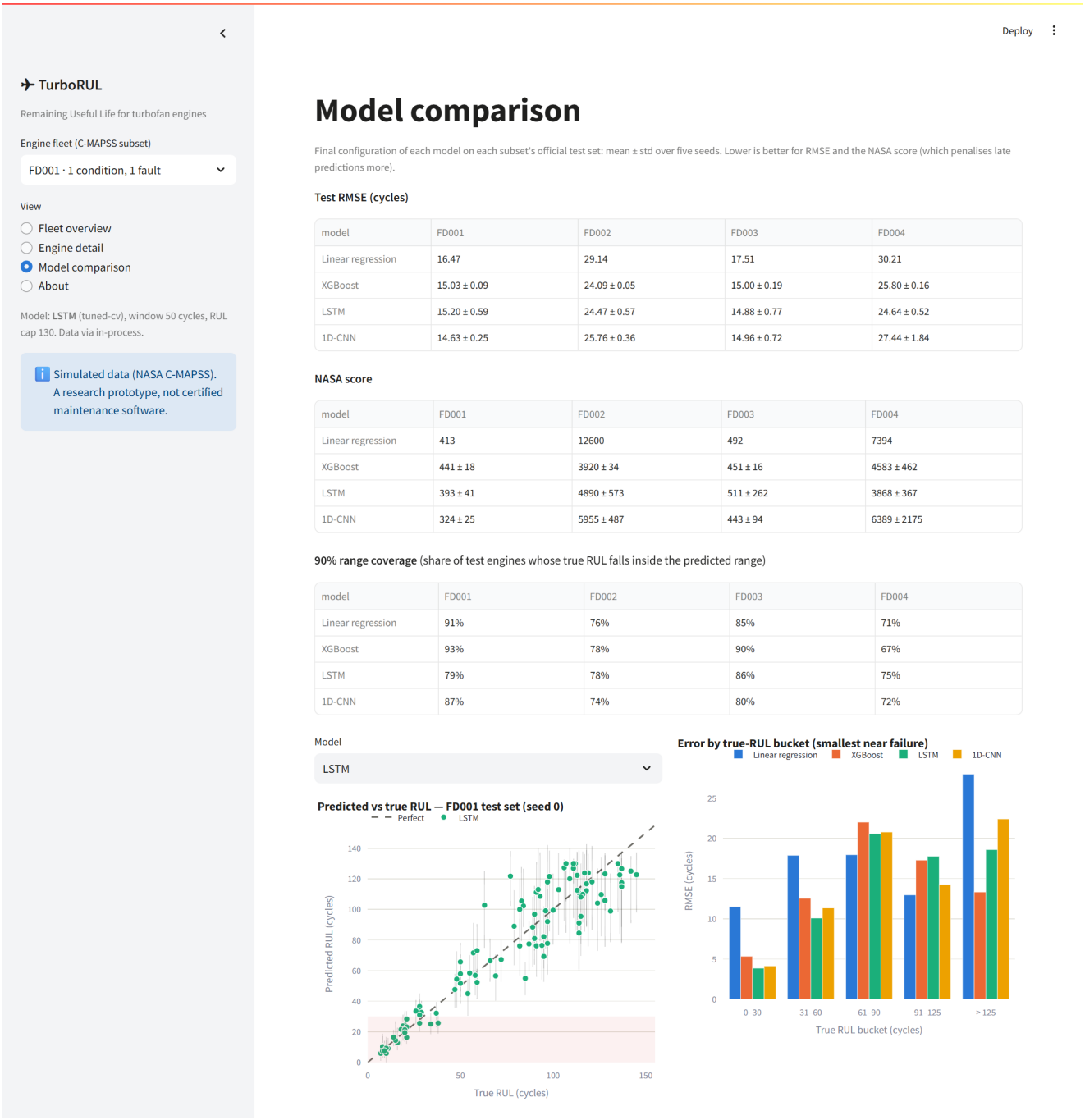


Figure 9. Model comparison: RMSE and NASA score per model and subset, range coverage, predicted vs true, error by RUL bucket.

8. Work completed (Phases 4–5)

Component	Purpose
split.py, pipeline.py, tune.py	Engine-level k-fold; fold-aware preparation; cross-validated Optuna tuning with fold-level pruning.
pipeline.RULModel	One interface for fit, predict, calibrated ranges (FR-8) and explanations (FR-10); saved with joblib.
explain.py, tiers.py	TreeSHAP, integrated gradients, exact linear contributions; tiers and actions (FR-9).
export.py, predict.py, service.py	Deployable bundles; prediction function (FR-7); JSON queries for API and dashboard.
api/, app/	FastAPI service and Streamlit dashboard (FR-11).
tests/	64 tests, all passing: api 6, dashboard 6, data 14, metrics 3, predict 5, preprocess 6, split 6, uncertainty_explain 12, windows 6.

9. Next steps (Phase 6)

- Deploy the dashboard (in-process mode) to Streamlit Community Cloud or Hugging Face Spaces; ship the four model bundles (small) and the test-engine data the fleet view needs.
- README with results table and demo GIF; course report; 3-minute demo video; write-up.
- Optional P2: replay mode (FR-12) — *predict_trajectory* already computes the prediction at every past cycle; fault-mode classification on FD003/FD004 (FR-13).
- Candidate improvements, judged on validation only: windows longer than 50 cycles (every tuned model chose the longest offered); per-subset LSTM tuning given more compute; an asymmetric loss to reduce late predictions.

10. Limitations

- The data is simulated; failure is a threshold crossing; there is no cost or schedule data, so maintenance value is argued, not measured.
- Ranges are calibrated on a few hundred truncated validation engines and against capped truth; they describe the model's target range, not lives beyond the cap.
- The LSTM on FD002–FD004 uses FD001's tuned configuration, so its comparison with the per-subset-tuned models is conservative.
- Test sets are small (100, 259, 100, 248 engines), so differences of a few tenths of a cycle are not meaningful, and FD001's test set has now been scored several times (descriptive use only).